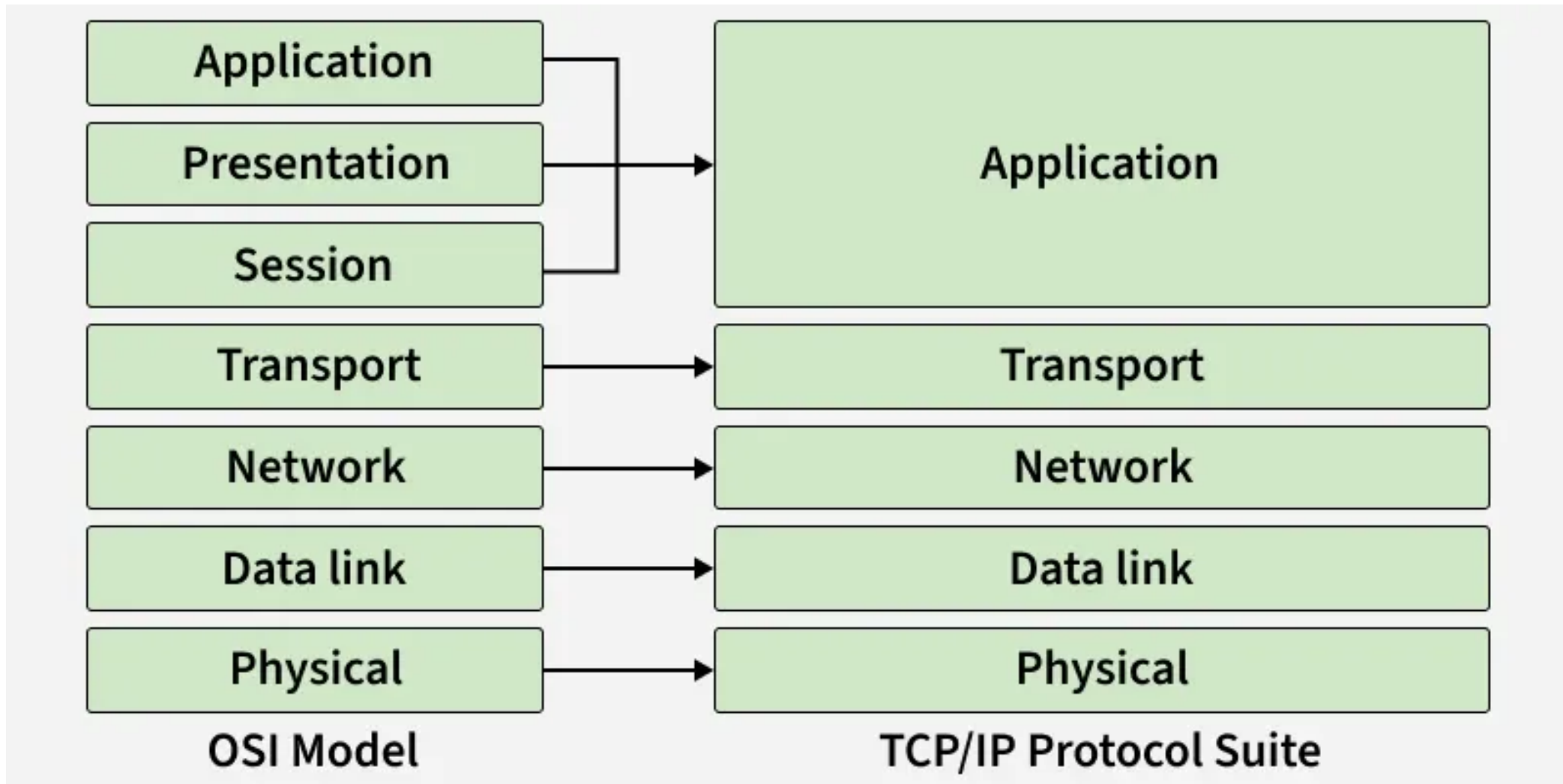


Unit -2

Computer Networks

TCP/IP MODEL

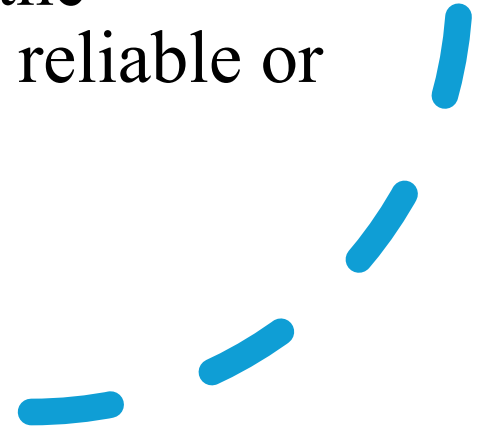


Application Layer

- ❖ The Application Layer is the top layer of the TCP/IP model and the one closest to the user. This is where all the apps you use like web browsers, email clients, or file sharing tools connect to the network.
- ❖ It acts like a bridge between your software (like Chrome, Gmail, or WhatsApp) and the lower layers of the network that actually send and receive data.
- ❖ It supports different protocols like HTTP (for websites), FTP (for file transfers), SMTP (for emails), and DNS (for finding website addresses).
- ❖ It also manages things like data formatting, so both sender and receiver understand the data, encryption to keep data safe, and session management to keep track of ongoing connections.

Transport Layer

- ❖ The Transport Layer is responsible for delivering data accurately, reliably, and in the correct order between two devices communicating over a network. Whenever you send something—like a message, file, video, or webpage request—the transport layer ensures it reaches the other side properly.
- ❖ This layer mainly uses two protocols: TCP and UDP, depending on whether the communication needs to be more reliable or more fast.



Network Layer

- ❖ The Internet Layer is used for finding the best path for data to travel across different networks so it can reach the right destination. It works like a traffic controller, helping data packets move from one network to another until they reach the correct device.
- ❖ This layer uses the Internet Protocol (IP) to give every device a unique IP address, which helps identify where data should go.
- ❖ The main job of this layer is routing deciding the best way for data to travel.
- ❖ It also takes care of packet forwarding (moving data from one point to another), fragmentation (breaking large data into smaller parts), and addressing.

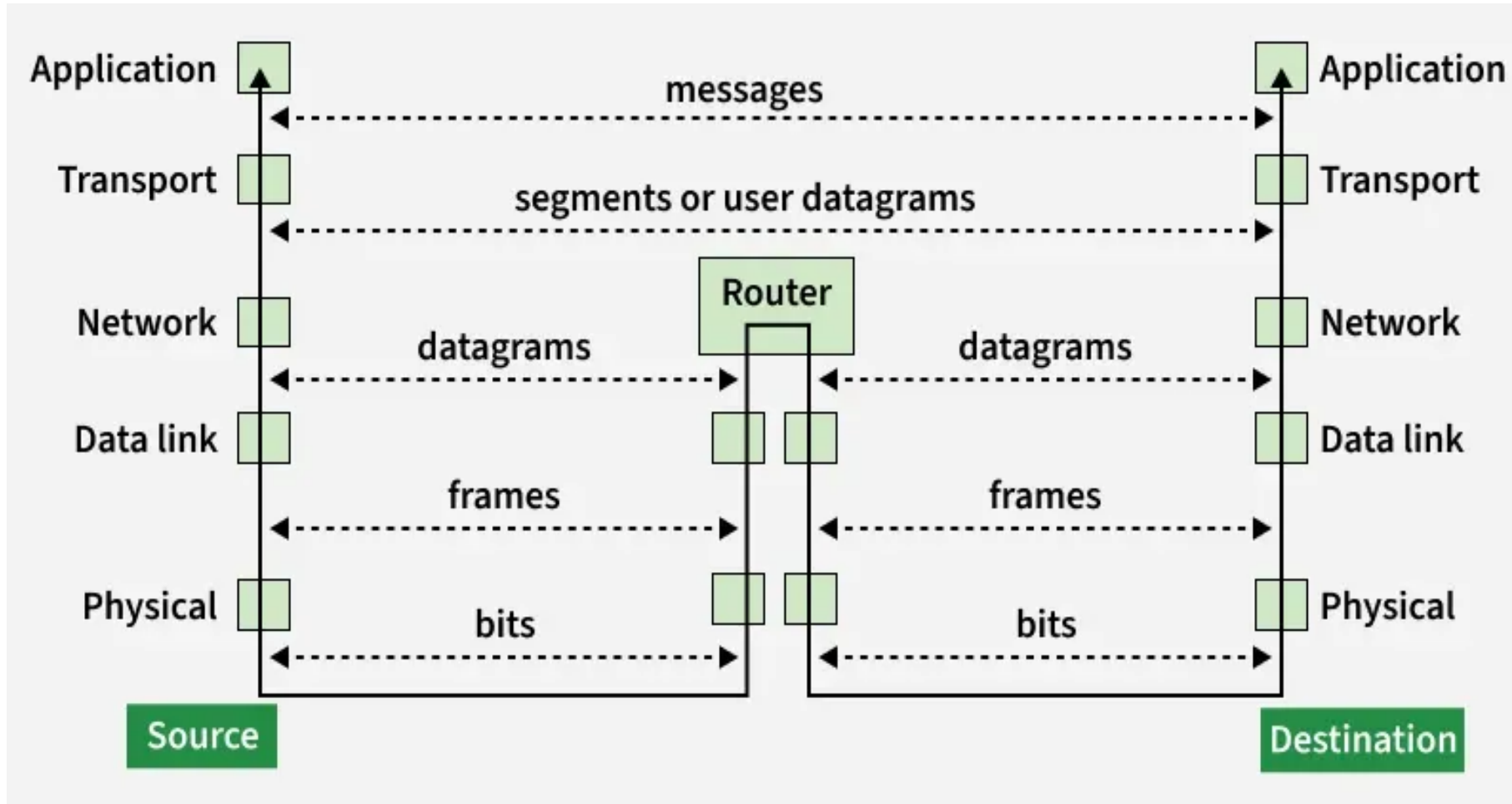
Data Link Layer

- ❖ The Internet is formed by multiple links (LANs & WANs) connected through routers.
- ❖ Routers decide the best path for datagrams to reach the destination.
- ❖ Different links may use different protocols at the data-link layer.
- ❖ TCP/IP does not specify any single data-link protocol — it supports all standard & proprietary ones.
- ❖ The data-link layer encapsulates the datagram into a frame for transmission.
- ❖ Link-layer services differ:
 - ❖ Some provide error detection & correction
 - ❖ Some provide only error detection
 - ❖ Others may offer minimal or no error control

Physical Layer

- ❖ Responsible for transmitting individual bits of a frame across a link.
- ❖ It is the lowest layer in the TCP/IP protocol suite.
- ❖ Communication at this layer is logical, because:
 - ❖ A transmission medium (like cable or air) exists below it.
- ❖ The transmission medium does not carry bits directly — it carries:
 - ❖ Electrical signals (wired media)
 - ❖ Optical signals (fiber-optic)
 - ❖ Radio waves (wireless)
- ❖ Bits from the data-link layer are converted into signals for transmission.

Working of TCP/IP Model



When Sending Data (From Sender to Receiver)

- ❖ **Application Layer:** Prepares user data using protocols like HTTP, FTP, or SMTP.
- ❖ **Transport Layer (TCP/UDP):** Breaks data into segments and ensures reliable (TCP) or fast (UDP) delivery.
- ❖ **Internet Layer (IP):** Adds IP addresses and decides the best route for each packet.
- ❖ **Link Layer (Network Access Layer):** Converts packets into frames and sends them over the physical network.

When Receiving Data (At the Destination)

- **Link Layer:** Receives bits from the network and rebuilds frames to pass to the next layer.
- **Internet Layer:** Checks the IP address, removes the IP header, and forwards data to the Transport Layer.
- **Transport Layer:** Reassembles segments, checks for errors, and ensures data is complete.
- **Application Layer:** Delivers the final data to the correct application (e.g., displays a web page in the browser).

Packet Transmission

- Packet transmission is the method of breaking down data into smaller units, called packets.
- This is essential for efficient and reliable delivery over a communication network.
- Each packet typically contains control information (headers, addresses, etc.) and payload (actual data).

Multiplexing

- Multiplexing is a technique used to transmit multiple signals over a single communication channel, optimizing resource utilization.
- **Types of Multiplexing:**
 - 1. Time Division Multiplexing (TDM):** Divides the channel into time slots for each signal.
 1. Example: Digital telephony systems.
 - 2. Frequency Division Multiplexing (FDM):** Assigns different frequency bands to each signal.
 1. Example: Radio broadcasting.
 - 3. Wavelength Division Multiplexing (WDM):** Used in optical networks, where each data stream is transmitted on a unique light wavelength.
 1. Example: Fiber optic communications.
 - 4. Code Division Multiplexing (CDM):** Uses unique codes to differentiate signals sharing the same frequency band.
 1. Example: Mobile cellular networks (CDMA).

Error Detection in Computer Networks

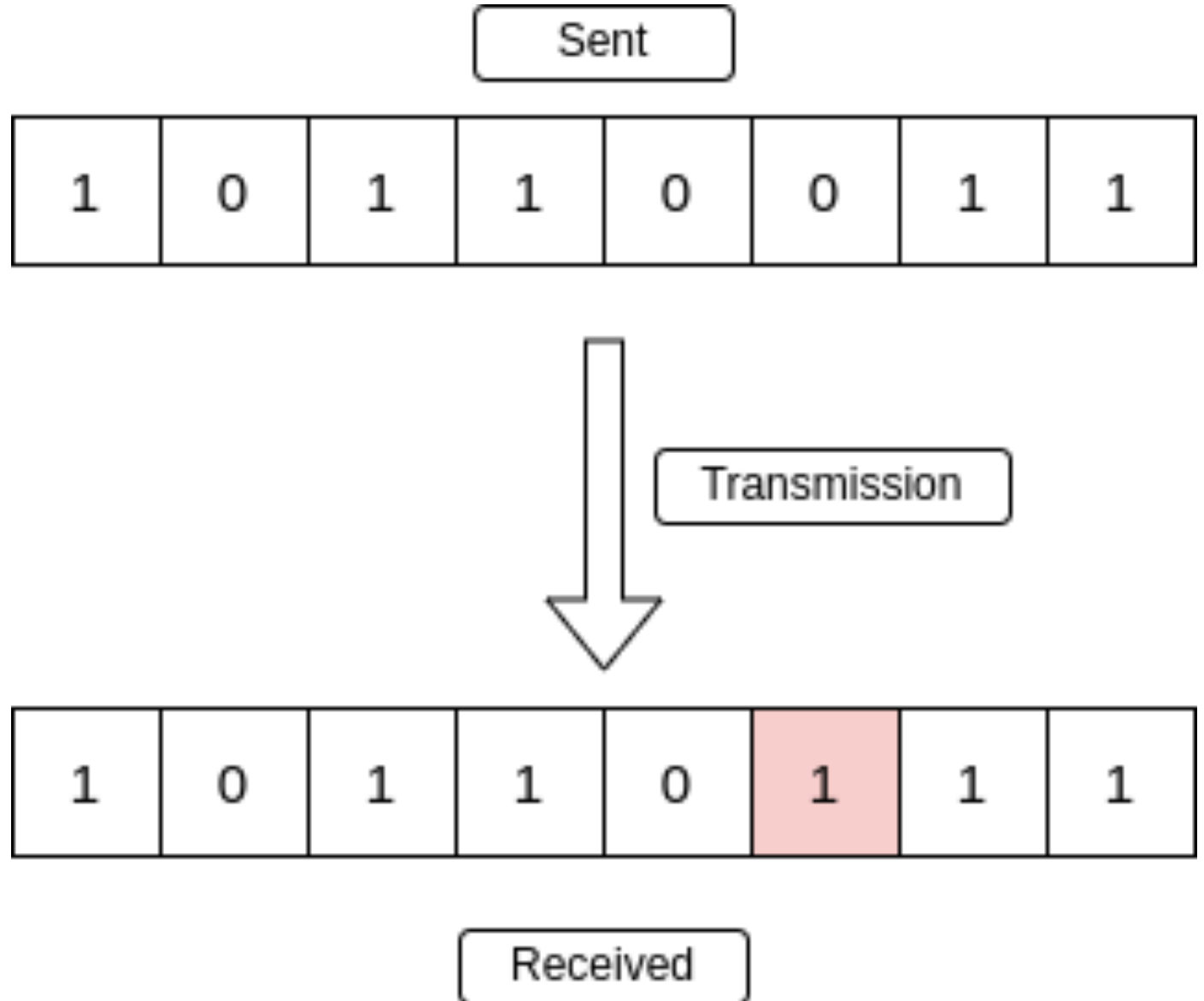
- Error is a condition when the receiver's information does not match the sender's.
- Digital signals suffer from noise during transmission that can introduce errors in the binary bits traveling from sender to receiver.
- That means a 0 bit may change to 1 or a 1 bit may change to 0.
- Data (Implemented either at the Data link layer or Transport Layer of the OSI Model) may get scrambled by noise or get corrupted whenever a message is transmitted.
- To prevent such errors, error-detection codes are added as extra data to digital messages.
- This helps in detecting any errors that may have occurred during message transmission.

Types of Errors

- Single-Bit Error
- Multiple-Bit Error
- Burst Error

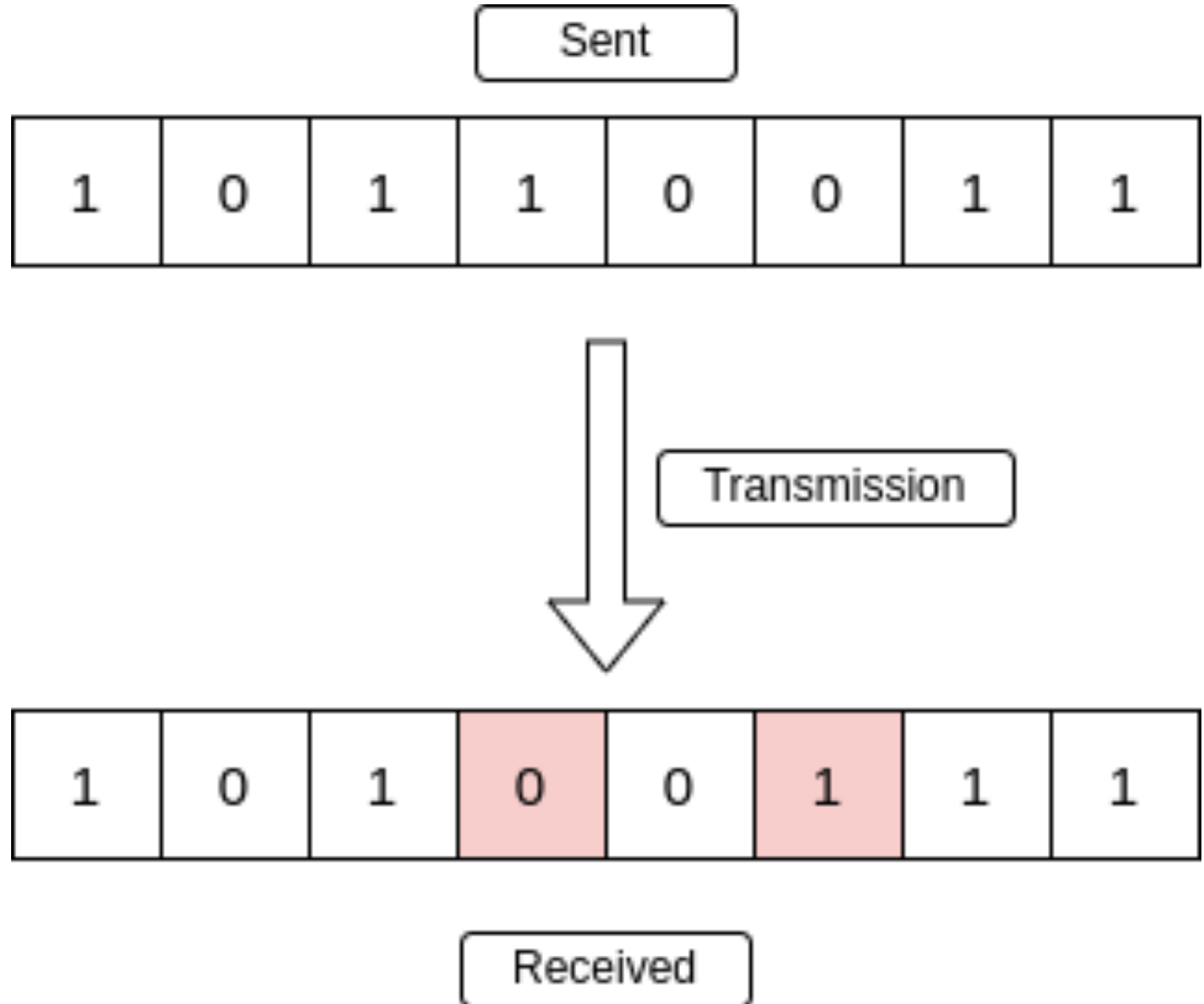
Single-Bit Error

- A single-bit error refers to a type of data transmission error that occurs when one bit (i.e., a single binary digit) of a transmitted data unit is altered during transmission, resulting in an incorrect or corrupted data unit.



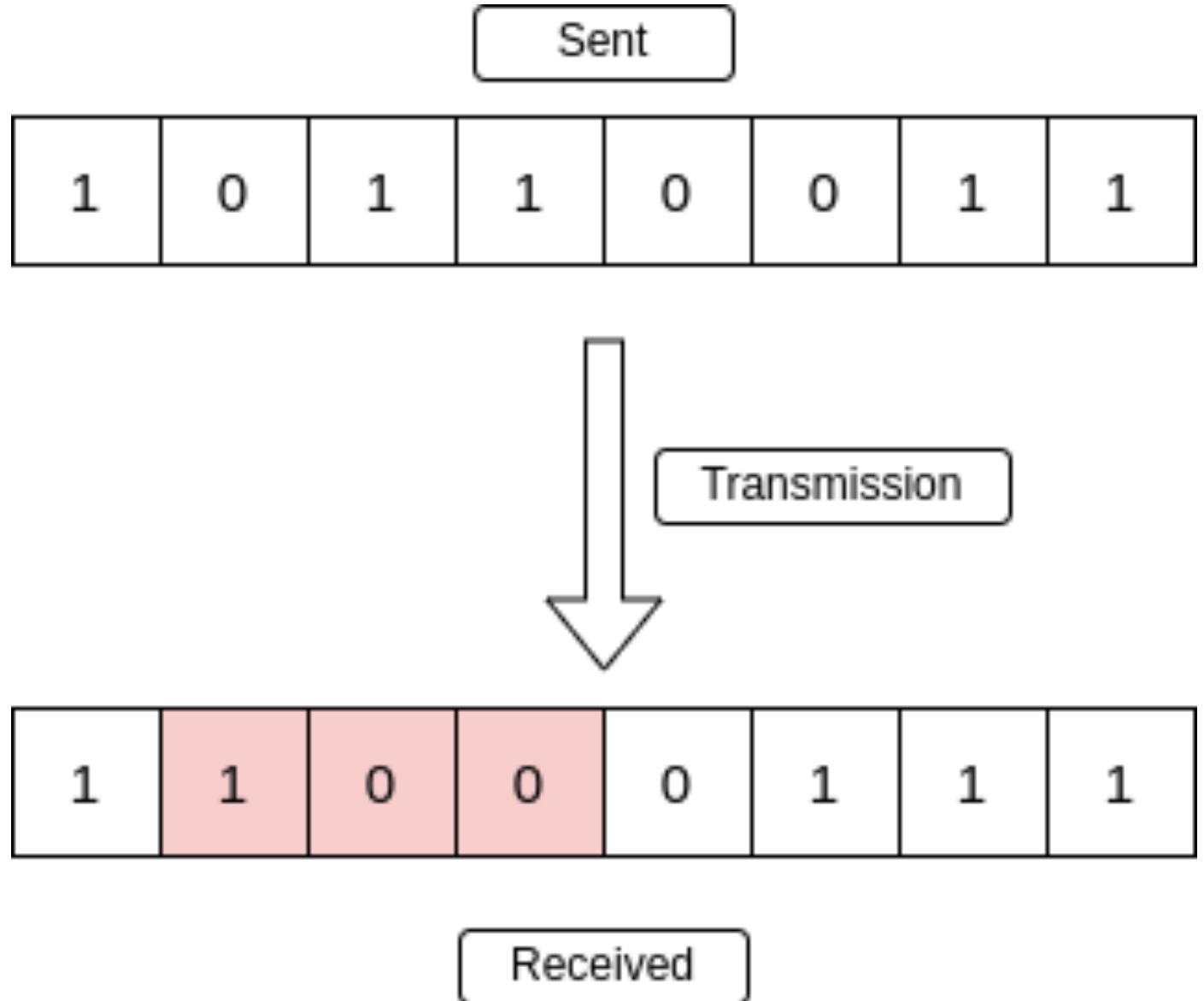
Multiple-Bit Error

- A multiple-bit error is an error type that arises when more than one bit in a data transmission is affected.
- Although multiple-bit errors are relatively rare when compared to single-bit errors, they can still occur, particularly in high-noise or high-interference digital environments.



Burst Error

- When several consecutive bits are flipped mistakenly in digital transmission, it creates a burst error. This error causes a sequence of consecutive incorrect values.



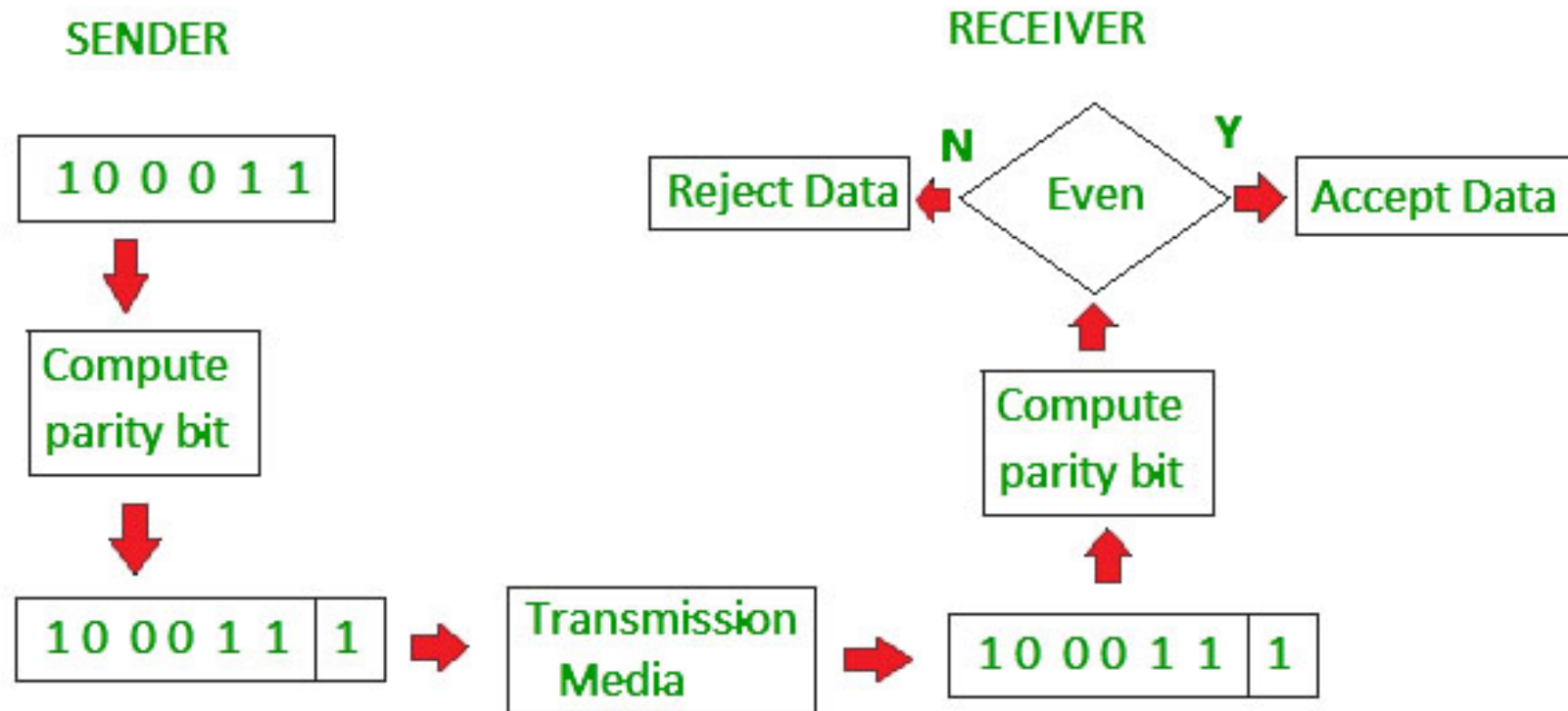
Error Detection Methods

- To detect errors, a common technique is to introduce redundancy bits that provide additional information.
- Various techniques for error detection include:
 - Simple Parity Check
 - Two-Dimensional Parity Check
 - Checksum
 - Cyclic Redundancy Check (CRC)

Simple Parity Check

- Simple-bit parity is a simple error detection method that involves adding an extra bit to a data transmission. It works as:
- 1 is added to the block if it contains an odd number of 1's, and
- 0 is added if it contains an even number of 1's

Simple Parity Check



Two-Dimensional Parity Check

Original Data

10011001	11100010	00100100	10000100
----------	----------	----------	----------

Row parities

10011001	0
11100010	0
00100100	0
10000100	0
11011011	0

Column
parities



100110010	111000100	001001000	100001000	110110110
-----------	-----------	-----------	-----------	-----------

Data to be sent

Two-Dimensional Parity Check

- Two-dimensional Parity check bits are calculated for each row, which is equivalent to a simple parity check bit.
- Parity check bits are also calculated for all columns, then both are sent along with the data.
- At the receiving end, these are compared with the parity bits calculated on the received data.

Checksum

- Checksum error detection is a method used to identify errors in transmitted data.
- The process involves dividing the data into equally sized segments and using a 1's complement to calculate the sum of these segments.
- The calculated sum is then sent along with the data to the receiver.
- At the receiver's end, the same process is repeated and if all zeroes are obtained in the sum, it means that the data is correct.

Checksum

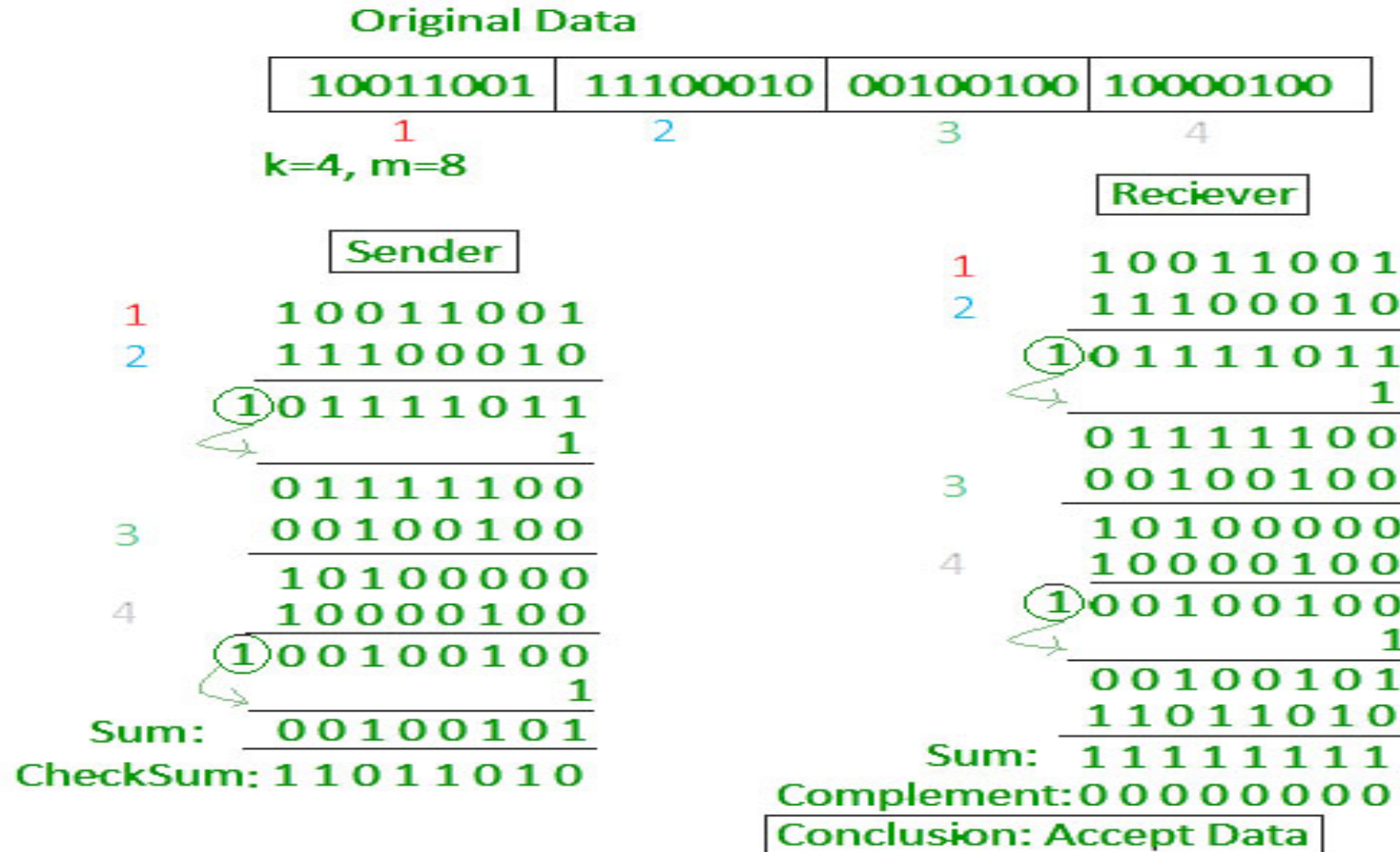
Checksum – Operation at Sender's Side

- Firstly, the data is divided into k segments each of m bits.
- On the sender's end, the segments are added using 1's complement arithmetic to get the sum. The sum is complemented to get the checksum.
- The checksum segment is sent along with the data segments.

Checksum – Operation at Receiver's Side

- At the receiver's end, all received segments are added using 1's complement arithmetic to get the sum. The sum is complemented.
- If the result is zero, the received data is accepted; otherwise discarded.

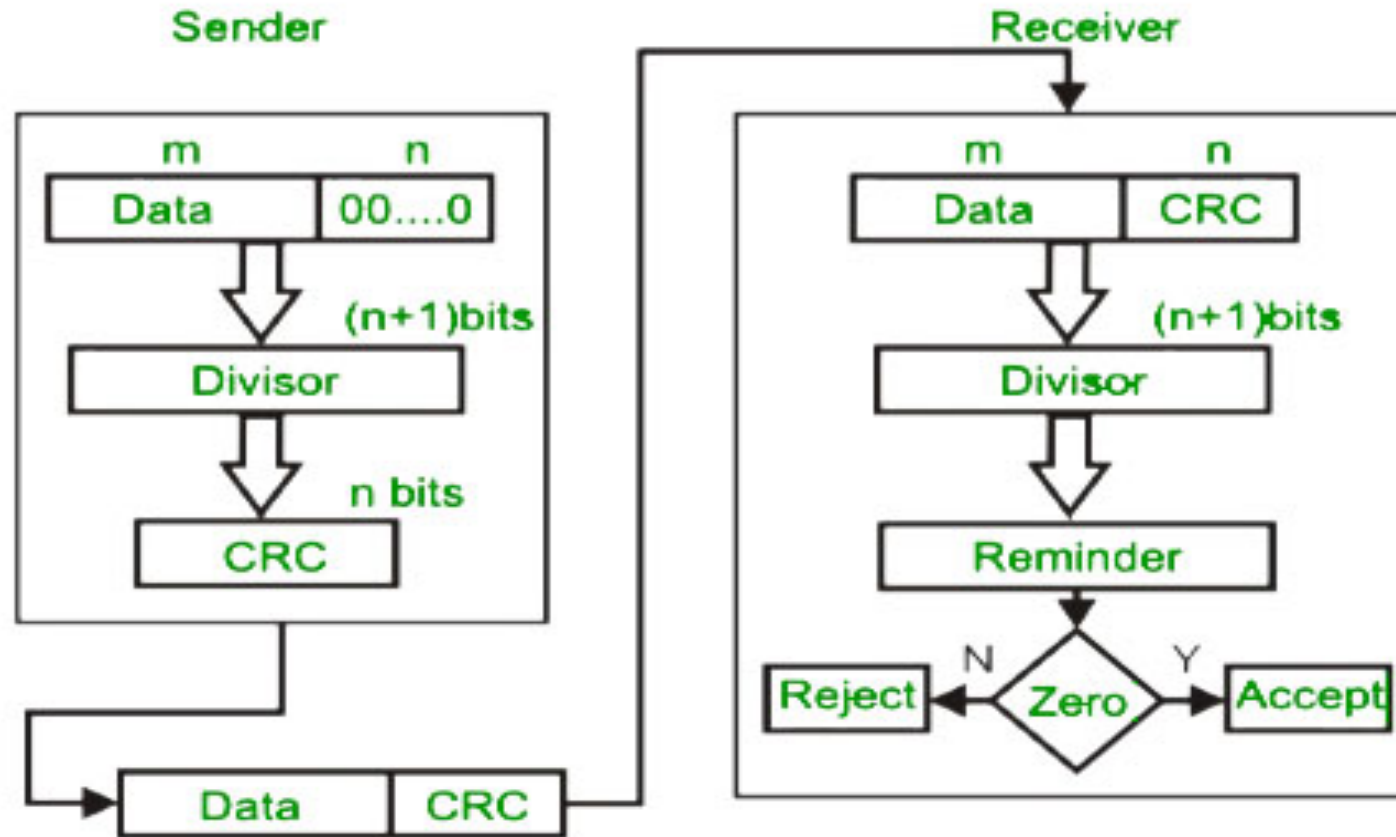
Checksum



Cyclic Redundancy Check (CRC)

- Unlike the checksum scheme, which is based on addition, CRC is based on binary division.
- In CRC, a sequence of redundant bits, called cyclic redundancy check bits, are appended to the end of the data unit so that the resulting data unit becomes exactly divisible by a second, predetermined binary number.
- At the destination, the incoming data unit is divided by the same number. If at this step there is no remainder, the data unit is assumed to be correct and is therefore accepted.
- A remainder indicates that the data unit has been damaged in transit and therefore must be rejected.

Cyclic Redundancy Check (CRC)



Cyclic Redundancy Check (CRC)

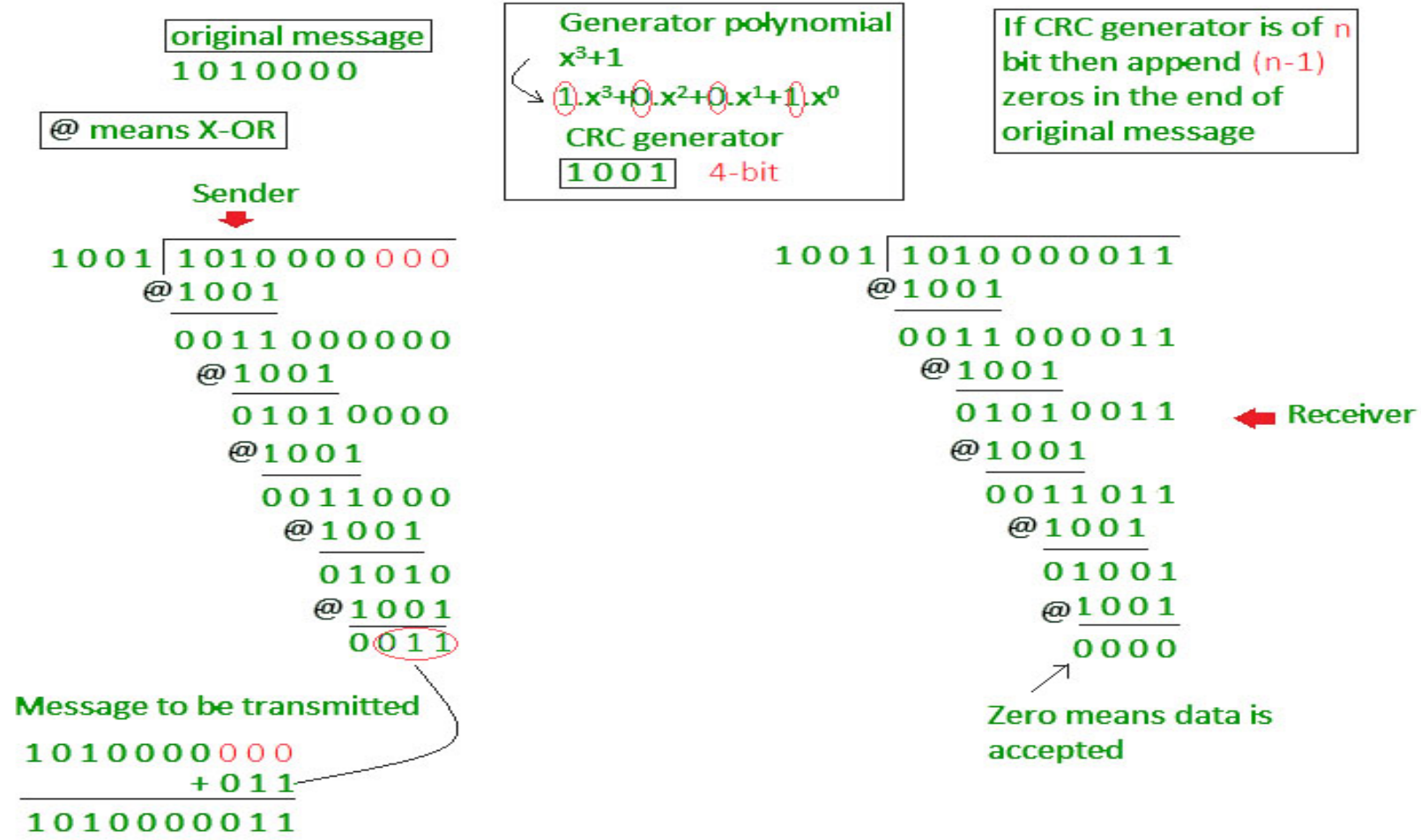
CRC Working

- We have given data word of length n and a divisor of length k .
- Step 1: Append $(k-1)$ zero's to the original message
- Step 2: Perform modulo 2 division
- Step 3: Remainder of division = CRC
- Step 4: Code word = Data with append $k-1$ zero's + CRC

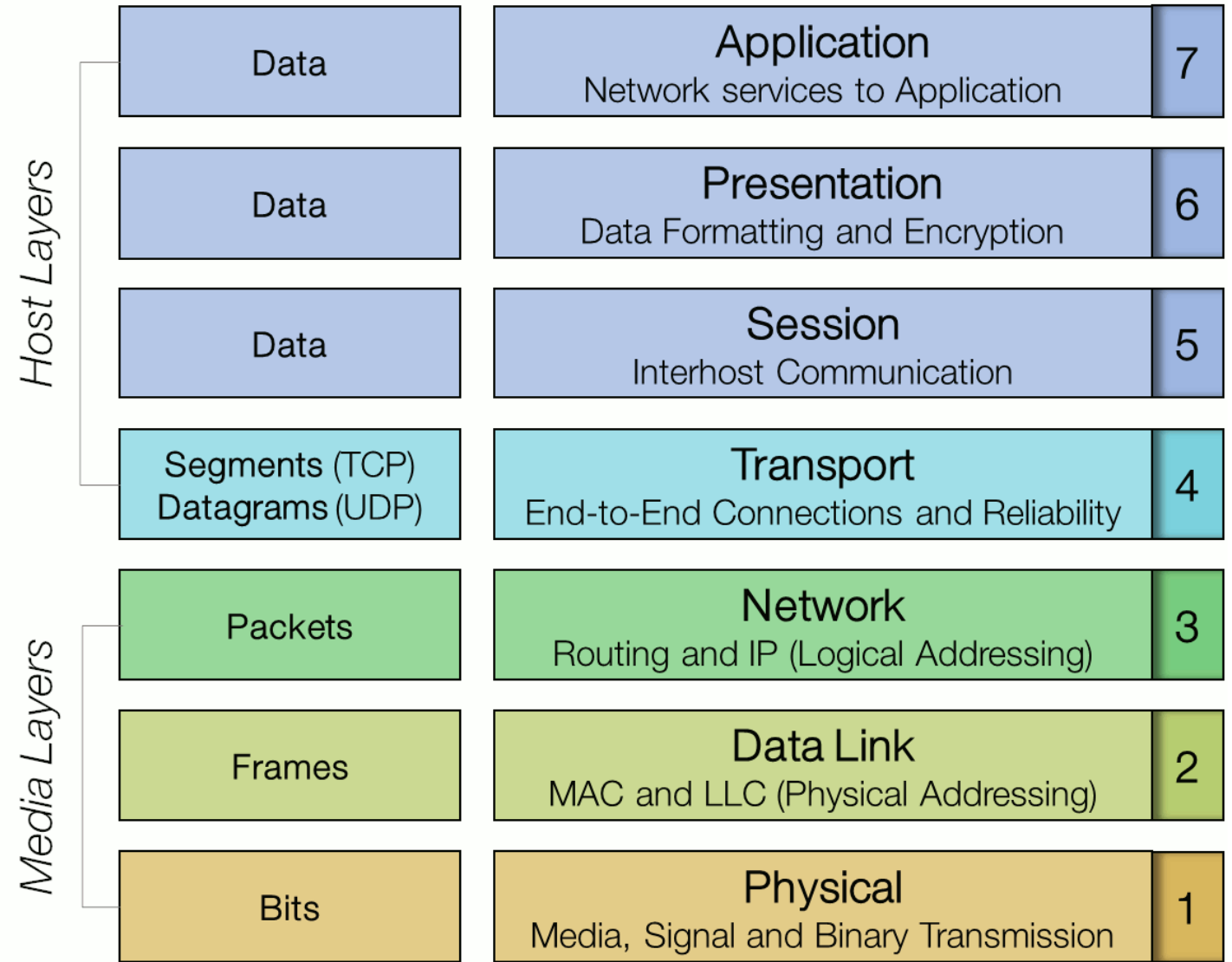
Note:

- CRC must be $k-1$ bits
- Length of Code word = $n+k-1$ bits

Cyclic Redundancy Check (CRC)



Flow of Data



Network Protocols

Network protocol is a set of rules that govern data communication between different devices in the network. It determines what is being communicated, how it is being communicated, and when it is being communicated. It permits connected devices to communicate with each other, irrespective of internal and structural differences.

Three major categories:

- ❖ Network Communication (HTTP, IP, TCP, UDP, DHCP)
- ❖ Network Management (Internet Control Message Protocol(ICMP), Simple Network Management Protocol(SNMP), FTP)
- ❖ Network Security(SSL, TLS, HTTPS)

LAN protocol

- LAN protocols enable communication within a local network.
- Data Link Layer protocols ensure reliable data transmission.
- Sliding Window Protocols improve transmission efficiency.

LAN protocol

- Ethernet (IEEE 802.3) – Uses CSMA/CD for collision detection.
- Wi-Fi (IEEE 802.11) – Uses CSMA/CA for wireless communication.
- Token Ring (IEEE 802.5) – Uses token-passing mechanism.
- FDDI – High-speed fiber-optic LAN protocol.

IEEE 802.3 and Ethernet

- Uses CSMA/CD (Carrier Sense Multiple Access with Collision Detection).
- Devices listen to the cable before sending data.
- If two devices transmit at the same time, a collision is detected and both retry.
- Common in wired LANs (homes, schools, offices).
- Modern switched Ethernet rarely experiences collisions.

Note: Full form : Institute of Electrical and Electronics Engineers used wired connection .

IEEE 802.11 Architecture and Services

- In the year 1990, IEEE 802.11 Committee formed a new working group, the IEEE 802.11 standard which defines protocols for Wireless Local Area Networks (WLANs).
- Just like how Ethernet provides services for wired media, IEEE 802.11 architecture is designed to provide features for wireless networks.
- Uses CSMA/CA (Carrier Sense Multiple Access with Collision Avoidance).
- Collisions are harder to detect wirelessly, so devices try to avoid them.
- Uses acknowledgments and waiting periods before transmitting. Used in wireless networks (phones, laptops, tablets).

Token Ring (IEEE 802.5)

- Uses a **token-passing mechanism**.
- A special frame called a **token** circulates around the network.
- Only the device holding the token can transmit → **no collisions**.
- Predictable but **slow and expensive**.

FDDI (Fiber Distributed Data Interface)

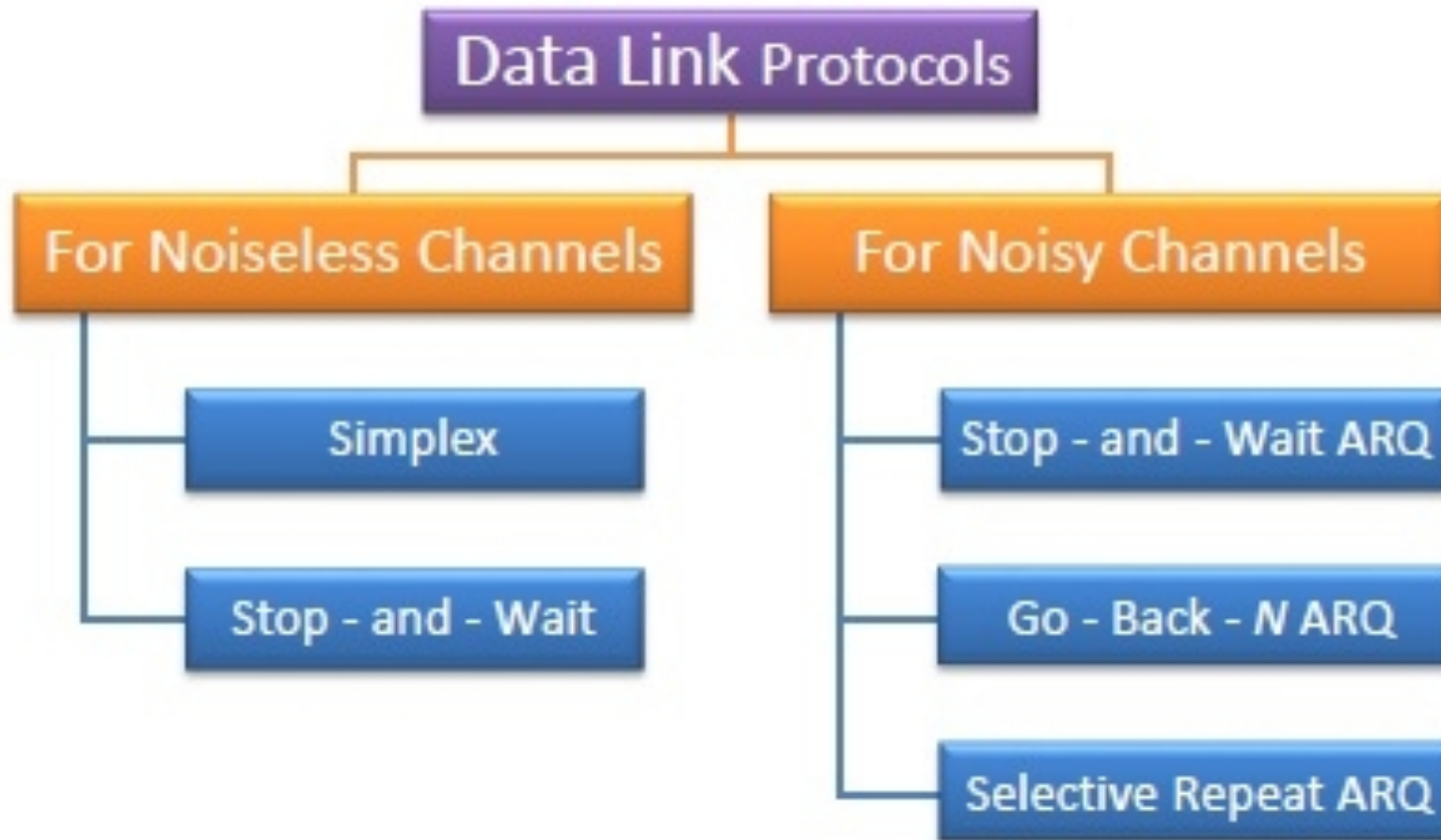
- A **high-speed fiber-optic LAN** technology.
- Uses **token passing** over fiber cables.
- Supports long distances and high reliability.
- Often used as a **backbone network**.
- Mostly replaced by **Fast Ethernet and Gigabit Ethernet**.

Technology	IEEE Standard	Transmission Medium	Access Method	Speed
Ethernet	IEEE 802.3	Twisted pair / Fiber	CSMA/CD	10 Mbps – 100+ Gbps
Wi-Fi	IEEE 802.11	Wireless (Radio waves)	CSMA/CA	11 Mbps – 9.6 Gbps
Token Ring	IEEE 802.5	Twisted pair	Token Passing	4–16 Mbps
FDDI	ANSI (related to IEEE)	Fiber-optic	Token Passing	100 Mbps

Elementary Data Link Protocols

- Protocols in the data link layer are designed so that this layer can perform its basic functions: framing, error control and flow control.
- Framing is the process of dividing bit - streams from physical layer into data frames whose size ranges from a few hundred to a few thousand bytes.
- Error control mechanisms deals with transmission errors and retransmission of corrupted and lost frames.
- Flow control regulates speed of delivery and so that a fast sender does not drown a slow receiver.

Types of Data Link Protocols



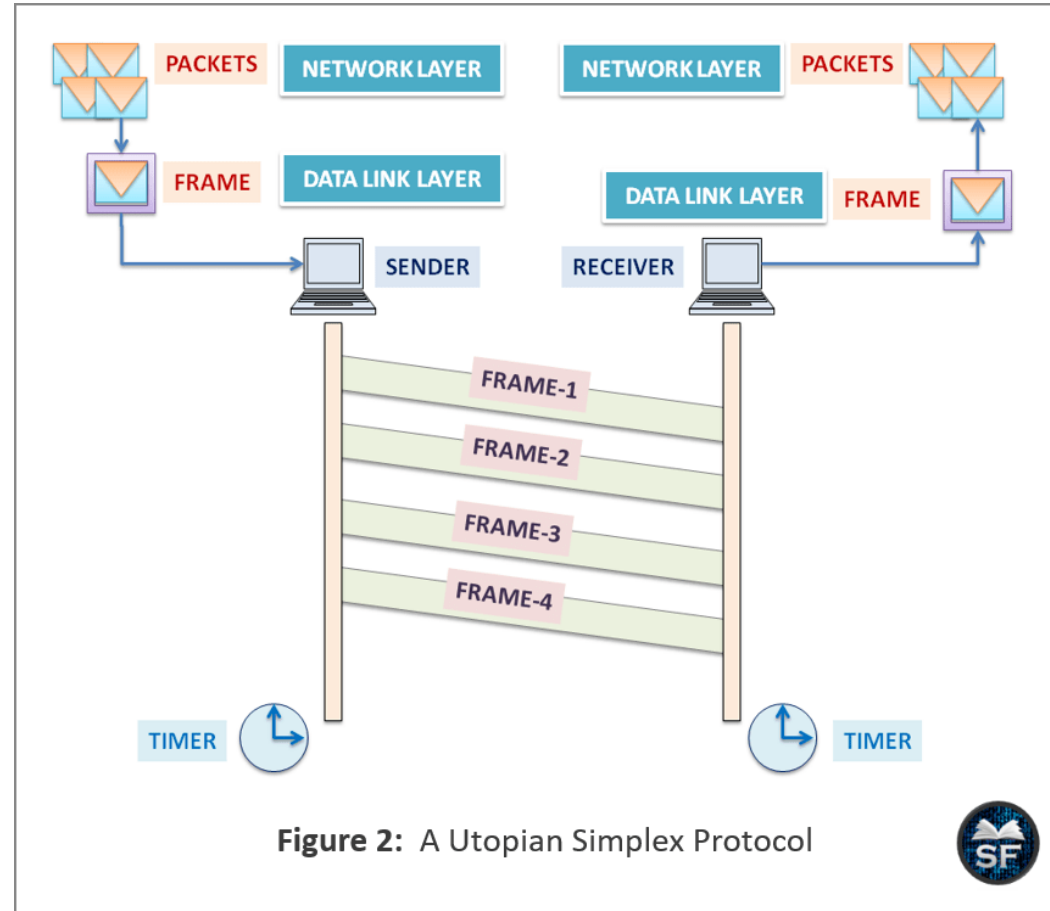
Simplex Protocol

- The Simplex protocol is a hypothetical protocol designed for unidirectional data transmission over an ideal channel, i.e. a channel through which transmission can never go wrong.
- The sender simply sends all its data available onto the channel as soon as they are available its buffer.
- The receiver is assumed to process all incoming data instantly. It is hypothetical since it does not handle flow control or error control.

Note:

- Assumes no errors and no flow control
- Sender sends frames continuously
- Receiver always ready to receive
- No acknowledgments

Simplex Protocol



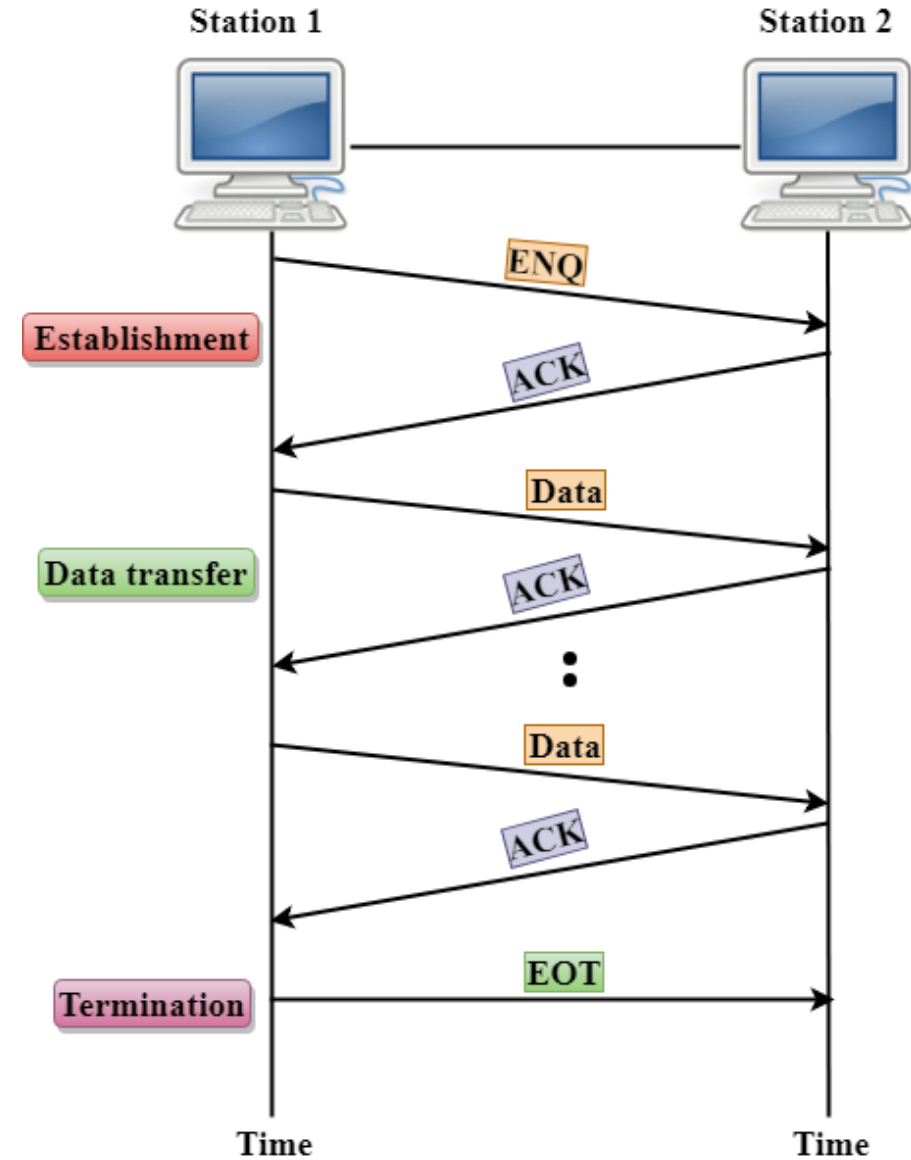
Stop – and – Wait Protocol

- Stop – and – Wait protocol is for noiseless channel too.
- It provides unidirectional data transmission without any error control facilities.
- The receiver has a finite buffer size with finite processing speed.
- The sender can send a frame only when it has received indication from the receiver that it is available for further data processing.
- Sender transmits one frame, waits for an acknowledgment.

Note:

- Sender sends one frame and waits
- Receiver sends acknowledgment (ACK)
- Prevents overflow at receiver
- No error handling required

Stop – and – Wait Protocol



Stop – and – Wait ARQ

- Stop – and – wait Automatic Repeat Request (Stop – and – Wait ARQ) is a variation of the above protocol with added error control mechanisms, appropriate for noisy channels.
- The sender keeps a copy of the sent frame. It then waits for a finite time to receive a positive acknowledgement from receiver.
- If the timer expires or a negative acknowledgement is received, the frame is retransmitted.
- If a positive acknowledgement is received then the next frame is sent.

Sliding Window Protocols

- Used for efficient data transmission.
- Sender maintains a window size (N) for unacknowledged frames.

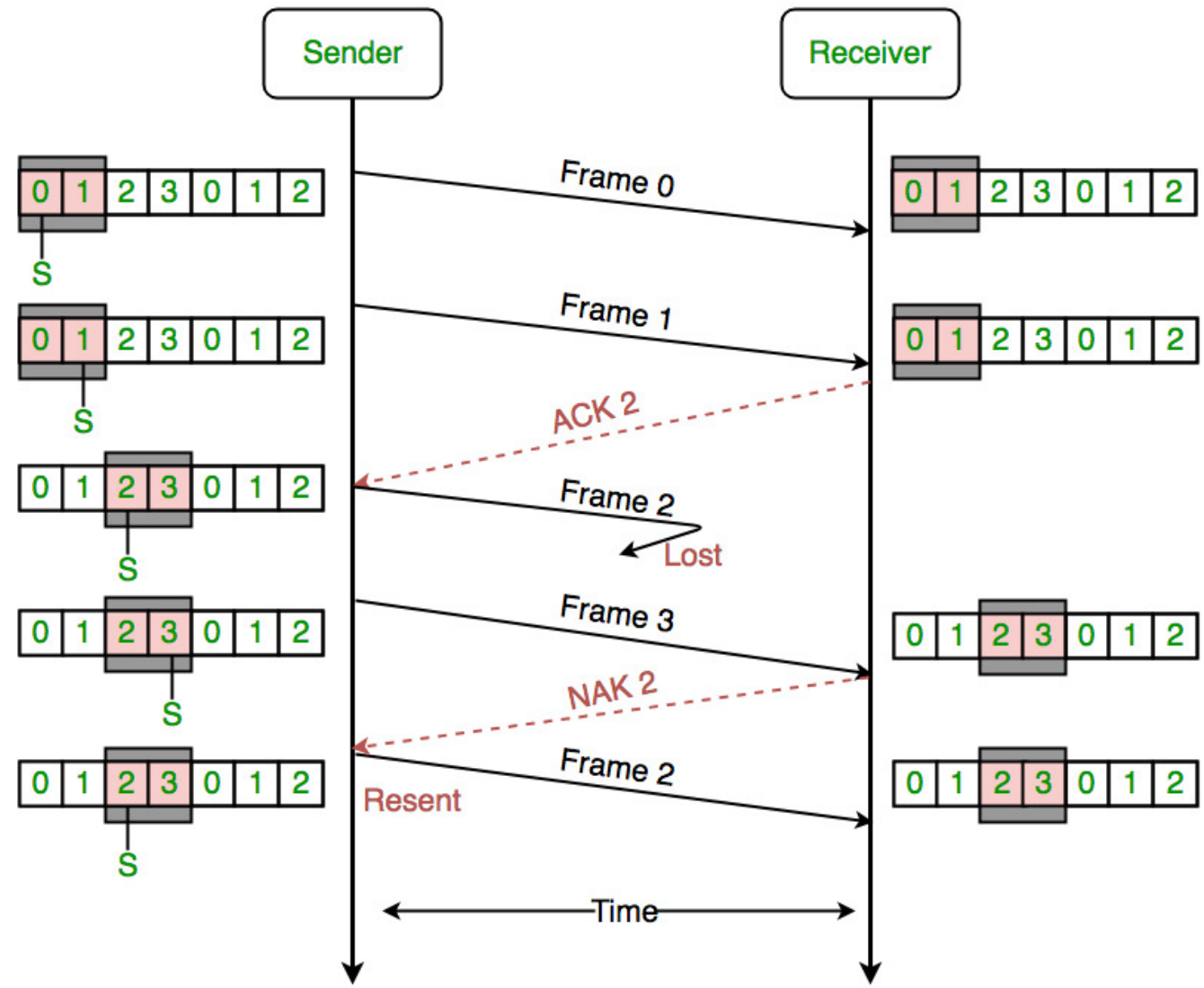
Types:

- Go-Back-N (GBN)
- Selective Repeat (SR)

Go – Back – N ARQ

- Go–Back – N ARQ provides for sending multiple frames before receiving the acknowledgment for the first frame.
- It uses the concept of sliding window, and so is also called sliding window protocol.
- The frames are sequentially numbered and a finite number of frames are sent. If the acknowledgement of a frame is not received within the time period, all frames starting from that frame are retransmitted.
- Sender can send up to N frames before stopping.
- If an error occurs, all frames from the error frame onward are retransmitted.
- Pros: Simple implementation.
- Cons: Wastes bandwidth due to unnecessary retransmissions.

Go – Back – N ARQ



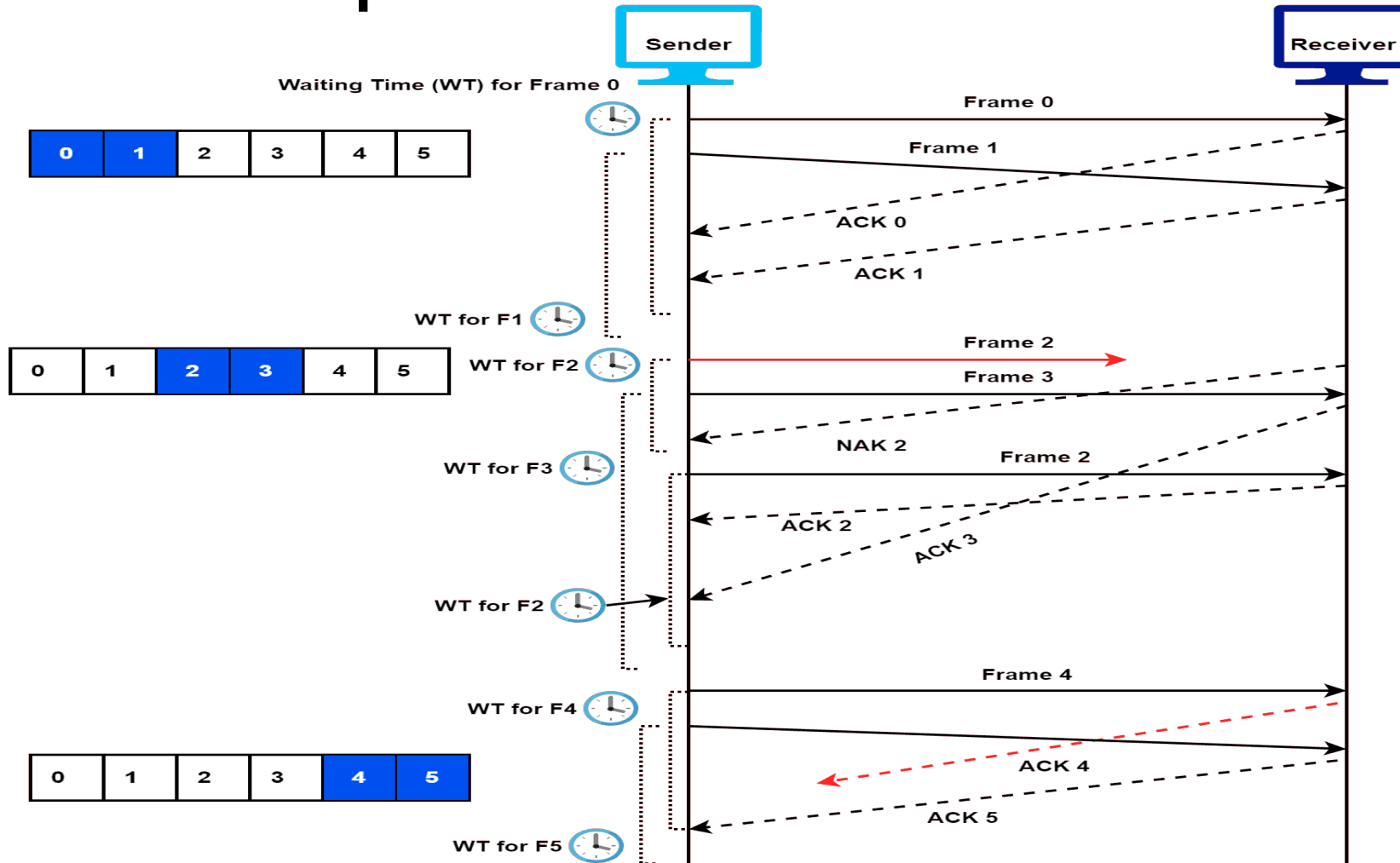
Selective Repeat ARQ

- This protocol also provides for sending multiple frames before receiving the acknowledgment for the first frame.
- However, here only the erroneous or lost frames are retransmitted, while the good frames are received and buffered.
- Only incorrect frames are retransmitted.
- Improves efficiency by avoiding redundant retransmissions.
- Pros: Efficient bandwidth usage.
- Cons: Requires additional buffer memory and complex processing.

Note: ACK → Acknowledgment

NAK → Negative Acknowledgment (received with errors / not received)

Selective Repeat ARQ



Conclusion

- LAN protocols define communication rules in a network.
- Data Link Layer protocols ensure reliable data transfer.
- Sliding Window Protocols improve efficiency by controlling frame transmission.
- Selective Repeat is more efficient than Go-Back-N but requires additional resources.